EXPERIMENT 18

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

K-Nearest Neighbors (KNN) Classification

Aim: Classify Iris flowers using the KNN algorithm.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
df=pd.read_csv('Iris.csv')
x=df.iloc[:,:-1]
y=df['variety']
xtr,xte,ytr,yte=train_test_split(x,y,test_size=0.2,random_state=1)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(xtr,ytr)
print("Accuracy:",model.score(xte,yte))
print("Prediction:",model.predict(xte[:5]))
```

Result

The KNN model predicts Iris species and displays test accuracy.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.